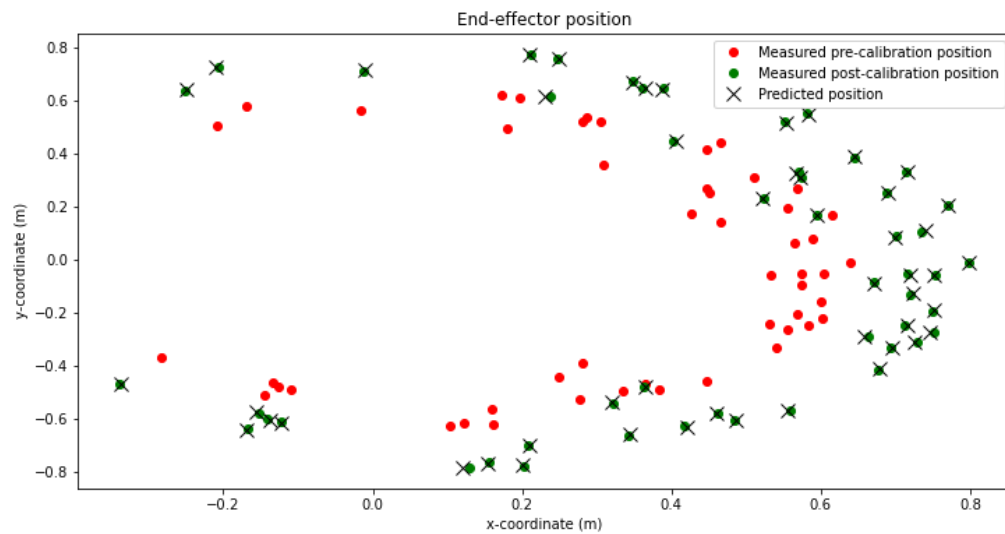


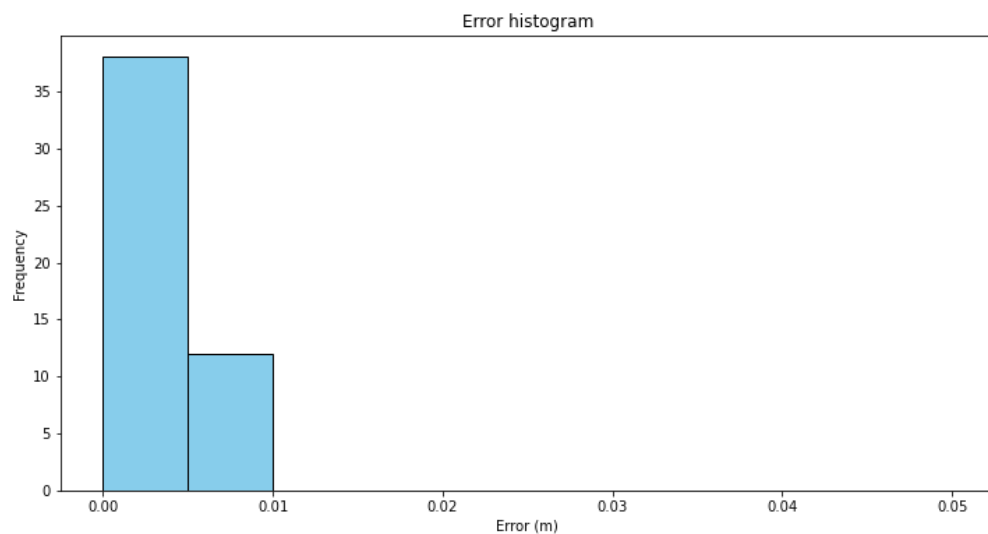
Results

A] Results:

1. The RMS error for parameters was reduced by 98.3% after their optimization.
2. The RMS error for end effector position was reduced by 97.2% after optimizing the parameters.
3. The measured pre-calibration, measured post-calibration and predicted positions of the end effector for all the different sets of joint angle pairs are shown in the graph below:



4. The measured post-calibration position of the end effector is more nearer to the predicted position than the measured pre-calibration position proving the effectiveness of the optimization.
5. The error distribution is shown in the form of a histogram below:



6. The histogram shows that most errors are closer to 0 m after the optimization process. The frequency of the bars of the histogram decreases as the error increases proving the effectiveness of the optimization of parameters.
7. The `scipy.optimize.least_squares` function was used to optimize the parameters because it is differentiable at every point including at 0. The function ensures that the parameters are optimized until the residuals function (error) reaches a minimum within specified lower and upper bounds. Inside those realistic bounds, the error is decreased to the lowest by adjusting the parameters.
8. The residuals function calculates the error between the measured and predicted values and returns an array of errors to the `least_squares` function for its next iteration.
9. The measured position of the end-effector progressively reaches its predicted position as the calibration advances, and in the process, compensates for the sensor errors too.